THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A train 150 m long crosses a platform 250 m long in 20 s. The speed of the train is: 1

(a) 12.5 m/s **(b)** 7.5 m/s **(c)** 8 m/s **(d)** 20 m/s

2. Two cyclists A and B start from the same point. A goes at 15 km/h and B at 20 km/h. After 3 h, B is ahead of A by: 1

(a) 60 km **(b)** 15 km **(c)** 5 km **(d)** 35 km

3. A pendulum was timed for 10 oscillations: 18.2 s, 18.0 s, 18.4 s. The best estimate of T is: 1

(a) 0.55 s **(b)** 1.82 s **(c)** 18.2 s **(d)** 1.8 s

4. A train moving at a steady 15 m/s runs for 20 s. The distance covered is: 1

(a) 35 m **(b)** 75 m **(c)** 0.75 m **(d)** 300 m

5. A train leaves at 6 a.m. and runs at a steady 60 km/h. By 12 noon it has covered: 1

(a) 120 km **(b)** 720 km **(c)** 60 km **(d)** 360 km

6. A sprinter's reaction time is 0.2 s and a stopwatch is started after seeing the start. The measured 10 s may be: 1

(a) double **(b)** more than true

(c) exact **(d)** less than true by about 0.2 s

7. A man runs round a circular track of length 400 m in 100 s. His average speed is: 1

(a) 400 m/s **(b)** 0 m/s **(c)** 40 m/s **(d)** 4 m/s

8. A man goes from A to B at 4 km/h and returns at 6 km/h. The time taken in the round trip is 5 h. The distance AB is: 1

(a) 24 km **(b)** 5 km **(c)** 12 km **(d)** 6 km

9. **Assertion (A):** The SI unit of time is the second. 1

Reason (R): One minute is equal to 60 seconds.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The time period of a simple pendulum increases when its length is increased. 1
Reason (R): A longer pendulum takes more time to complete one oscillation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the speed of a train in km/h if it crosses a 300 m bridge in 30 s and its length is 200 m. 2

12. Why can't the average speed of a round trip be the average of two speeds? 2

13. Why can we use speed = distance/time directly for uniform motion? 2

14. A student recorded: 10 oscillations in 15.2 s, 15.0 s, 15.4 s. Find the mean time period. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A train 120 m long crosses a pole in 12 s and a platform in 30 s. Find its speed and the length of the platform. 3

16. A bus leaves a town at 5 a.m. and runs at 40 km/h until 8 a.m. Find the distance covered. 3

17. The speed of sound is 340 m/s. How long does the sound take to travel 1.7 km? 3

OR

A person goes from A to B at 40 km/h and returns at 60 km/h. Find the average speed. 3

18. Explain the use of the 'seconds pendulum' in old clocks. 3

OR

A car's speed is 90 km/h. How far does it go in 1 s in metres? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Kavya makes a simple pendulum using a stone tied to a 1 m thread. She counts 30 complete oscillations in 60 seconds. Later, she makes a longer pendulum and finds it takes 50 seconds for 20 oscillations. 5
(i) Find the time period of the first pendulum.
(ii) Does the time period depend on the mass of the bob?
(iii) Find the time period of the longer pendulum.
(iv) Which pendulum is longer, and how do you know?

20. Design an experiment to show how the time period of a pendulum varies with its length. Specify measurements, tables and expected relation. 5

OR

A race track is 400 m. A runner completes 5 laps in 10 minutes. Find the speed in m/s and km/h. 5

END OF QUESTION PAPER — Total: 40 Marks